



Filter Wizard

Filter Wizard Design

Created on 05/29/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 2nd order Butterworth

Specifications: Optimize: Power; +Vs: 5; -Vs: -5

Gain: 0 dB

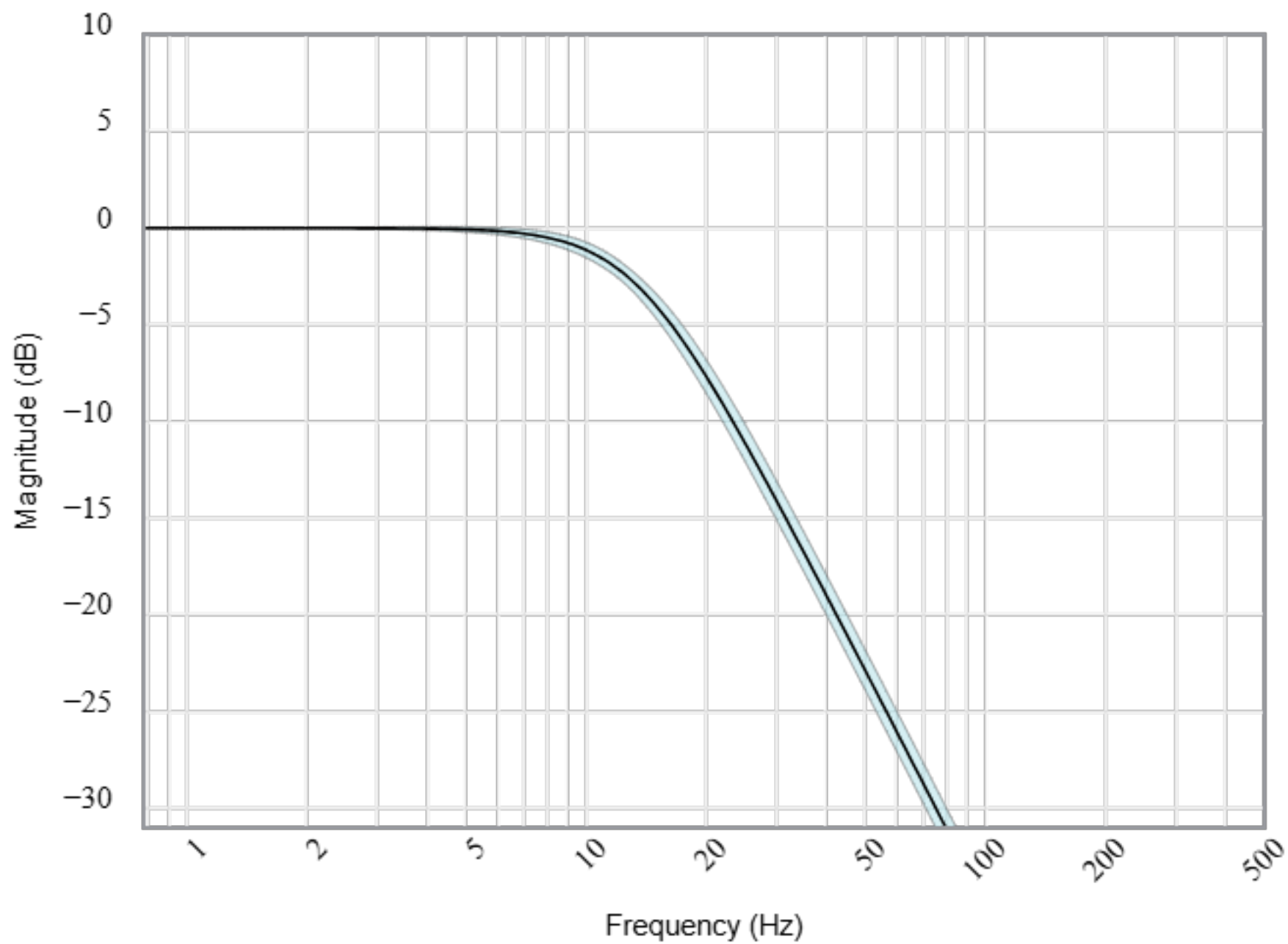
Passband: -0.5dB at 8Hz

Stopband: -11dB at 50Hz

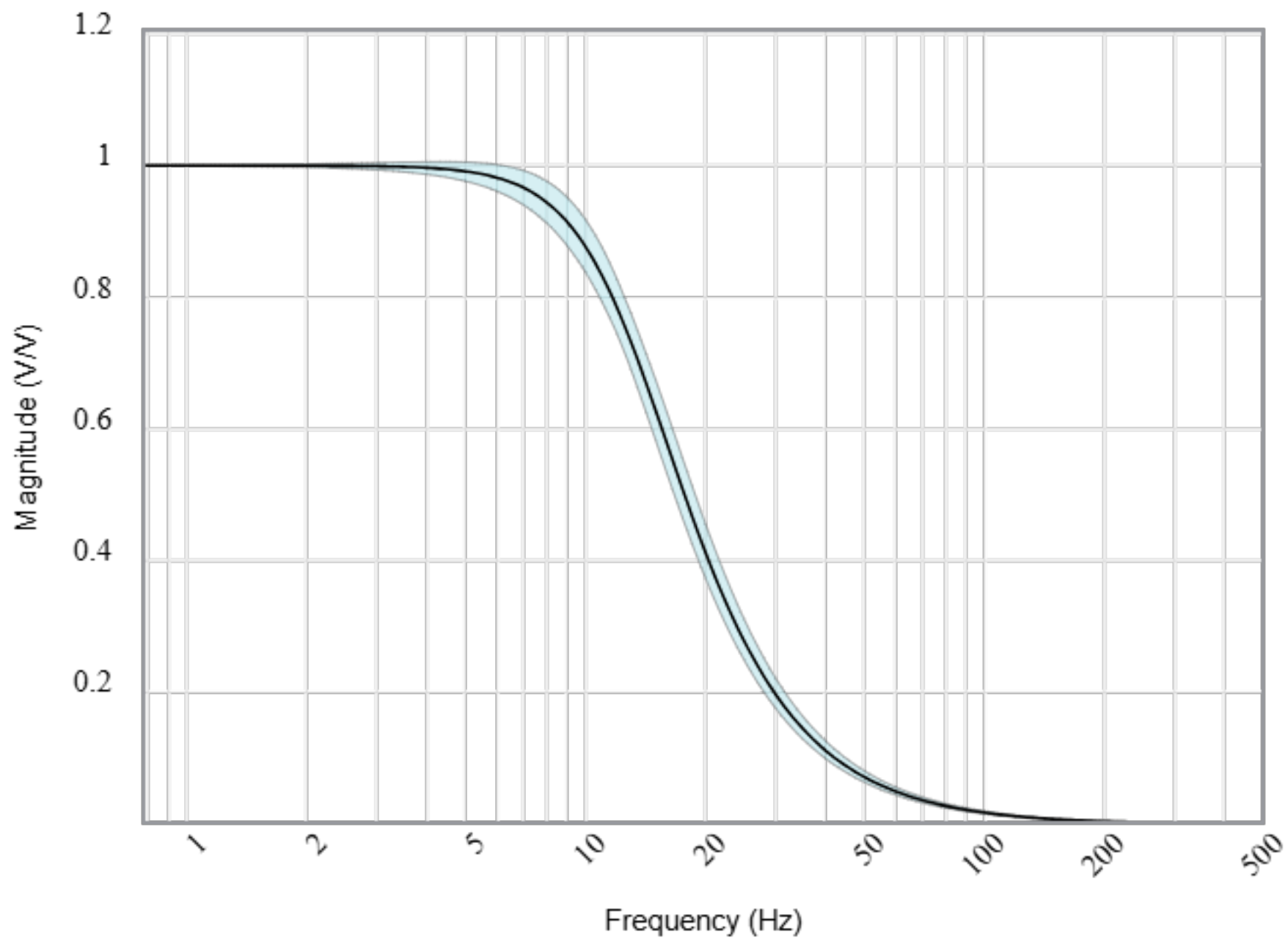
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

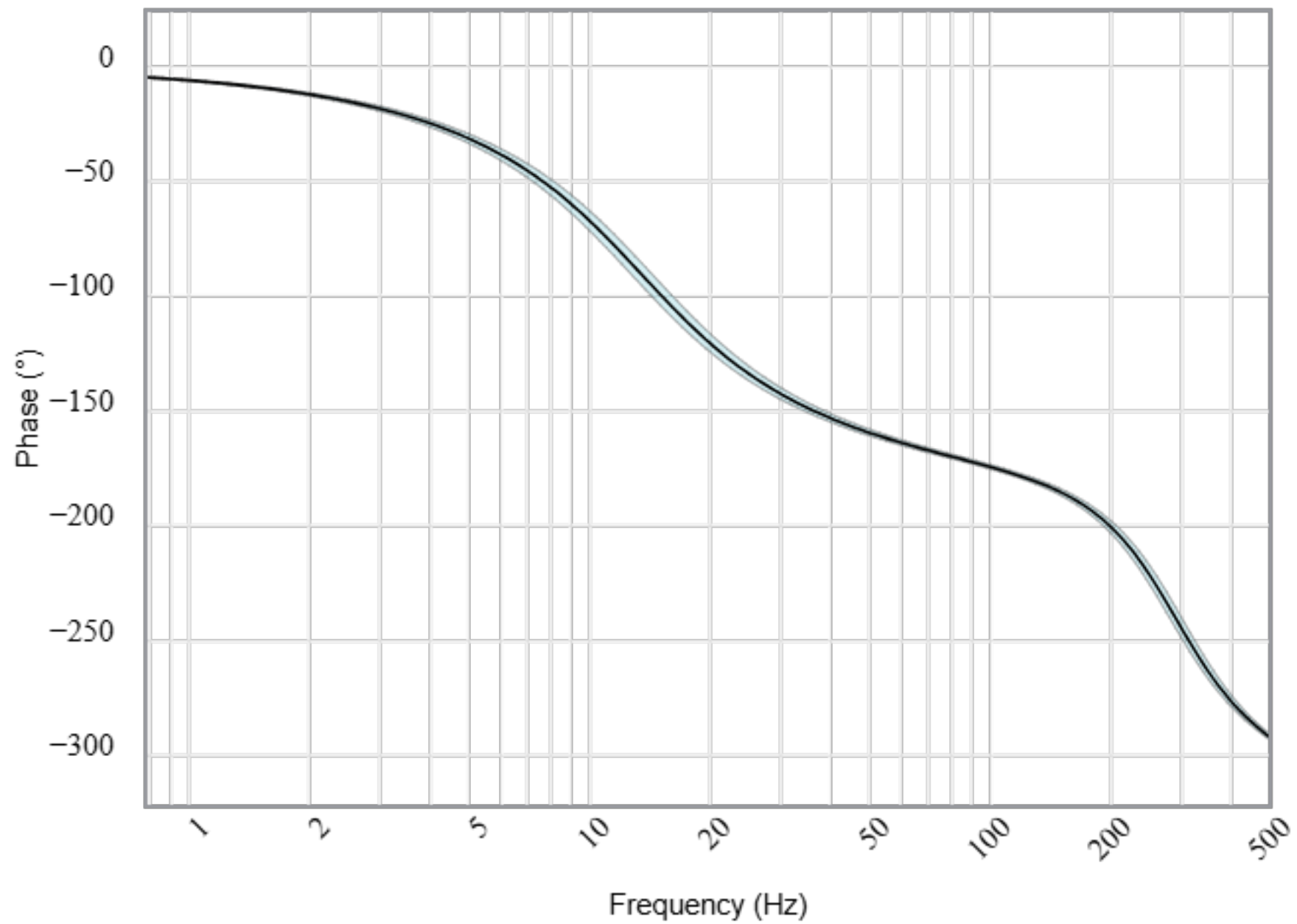
Magnitude(dB)



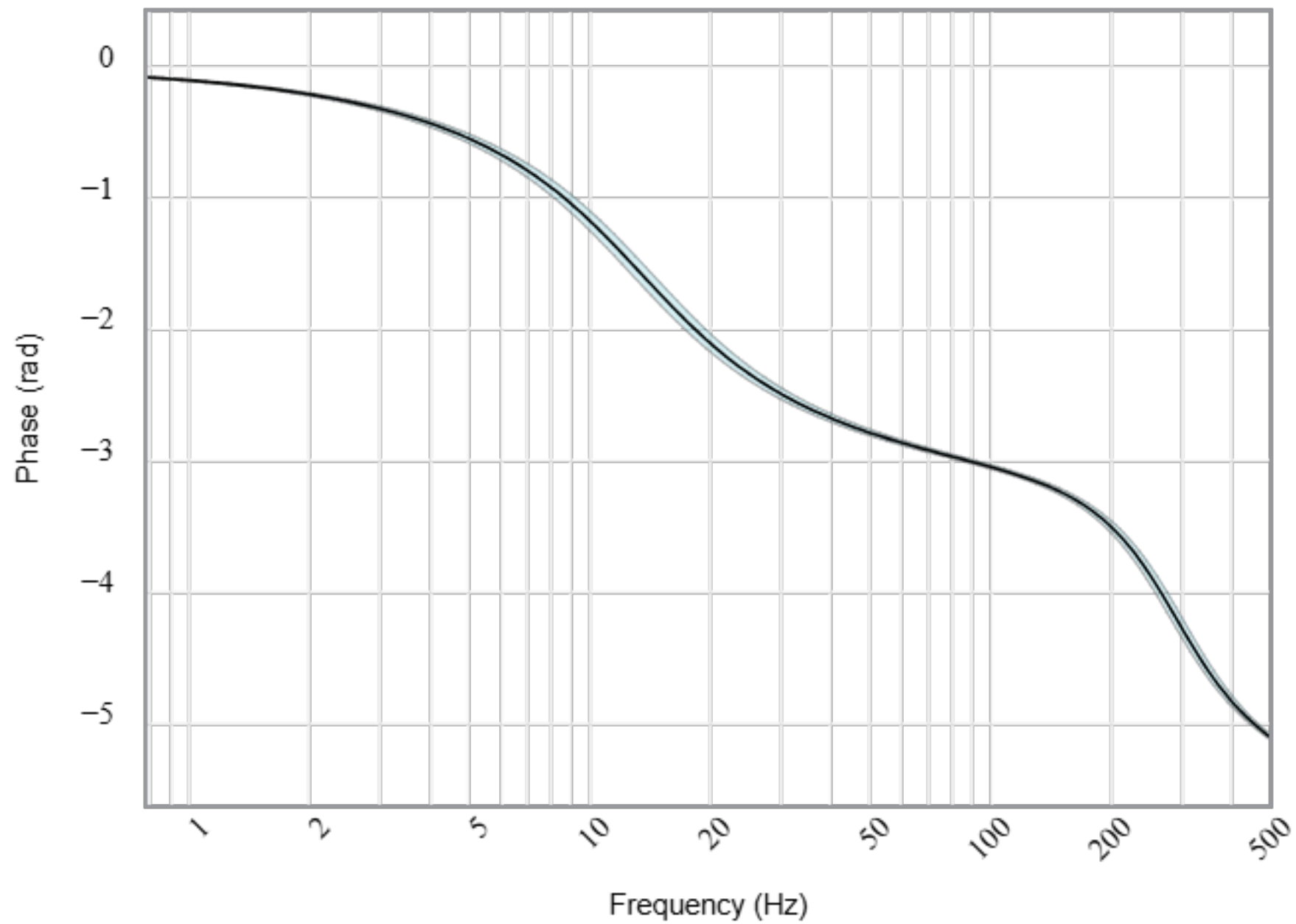
Magnitude(Volts per Volt)



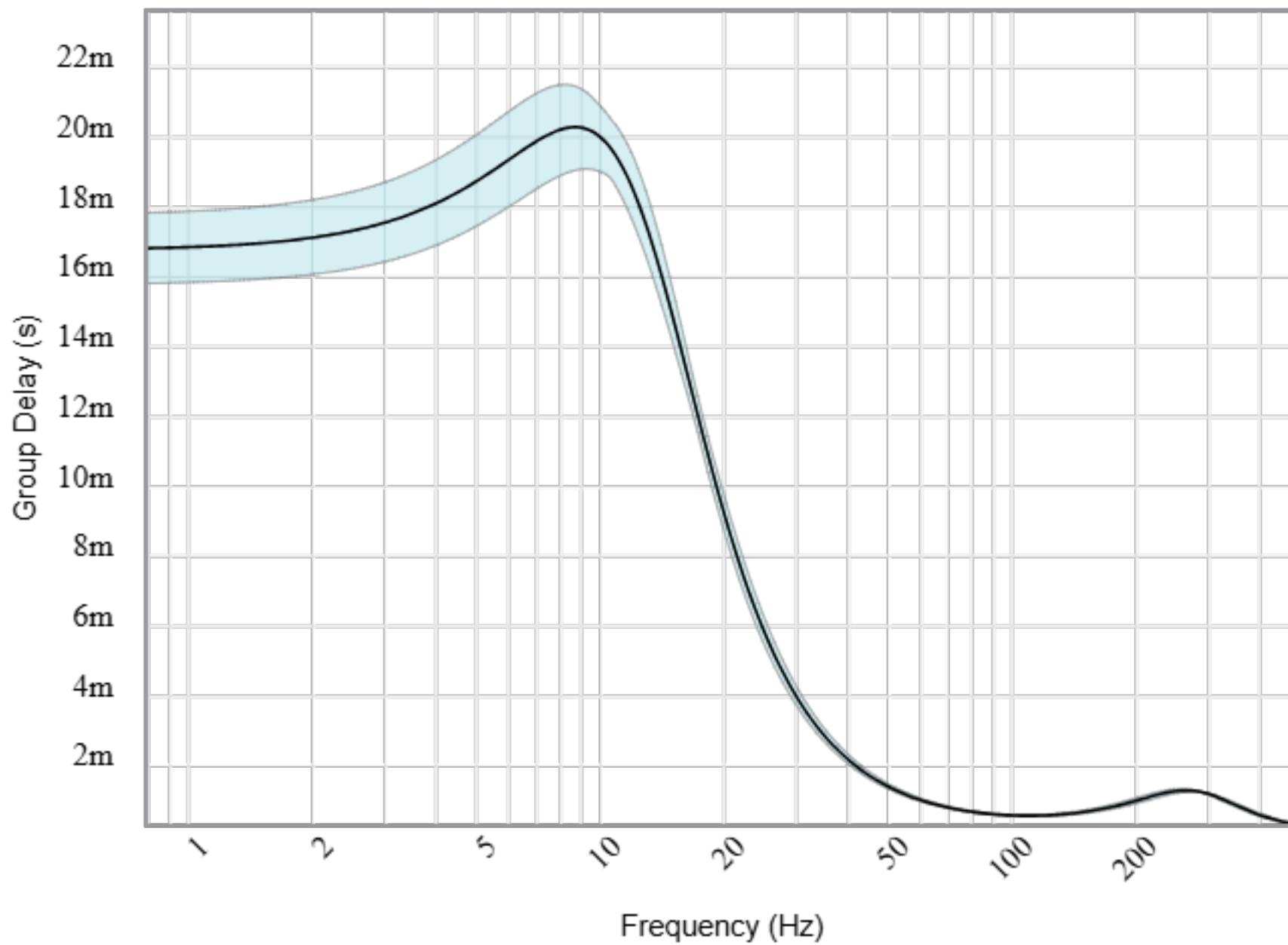
Phase(degrees)



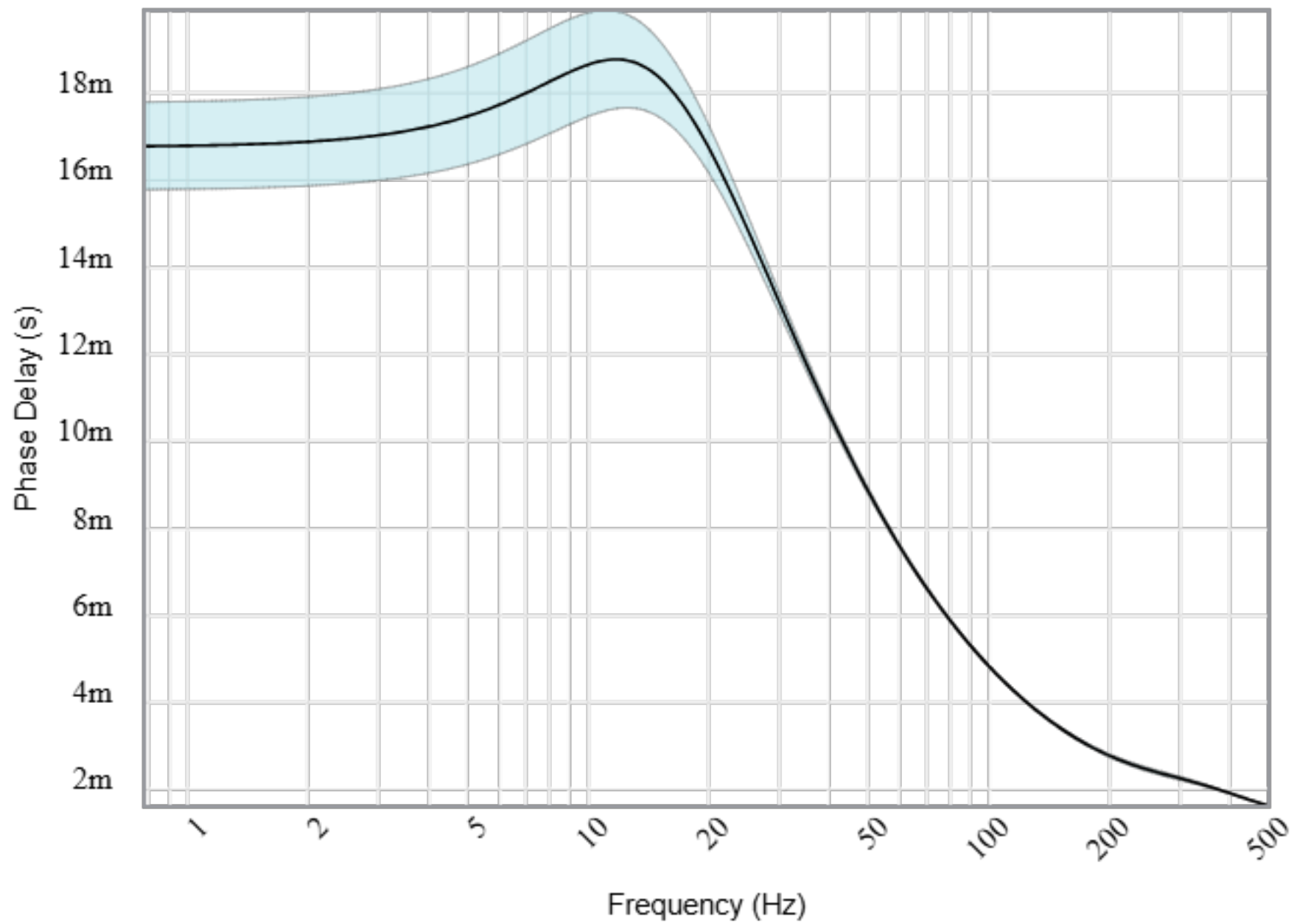
Phase(radians)



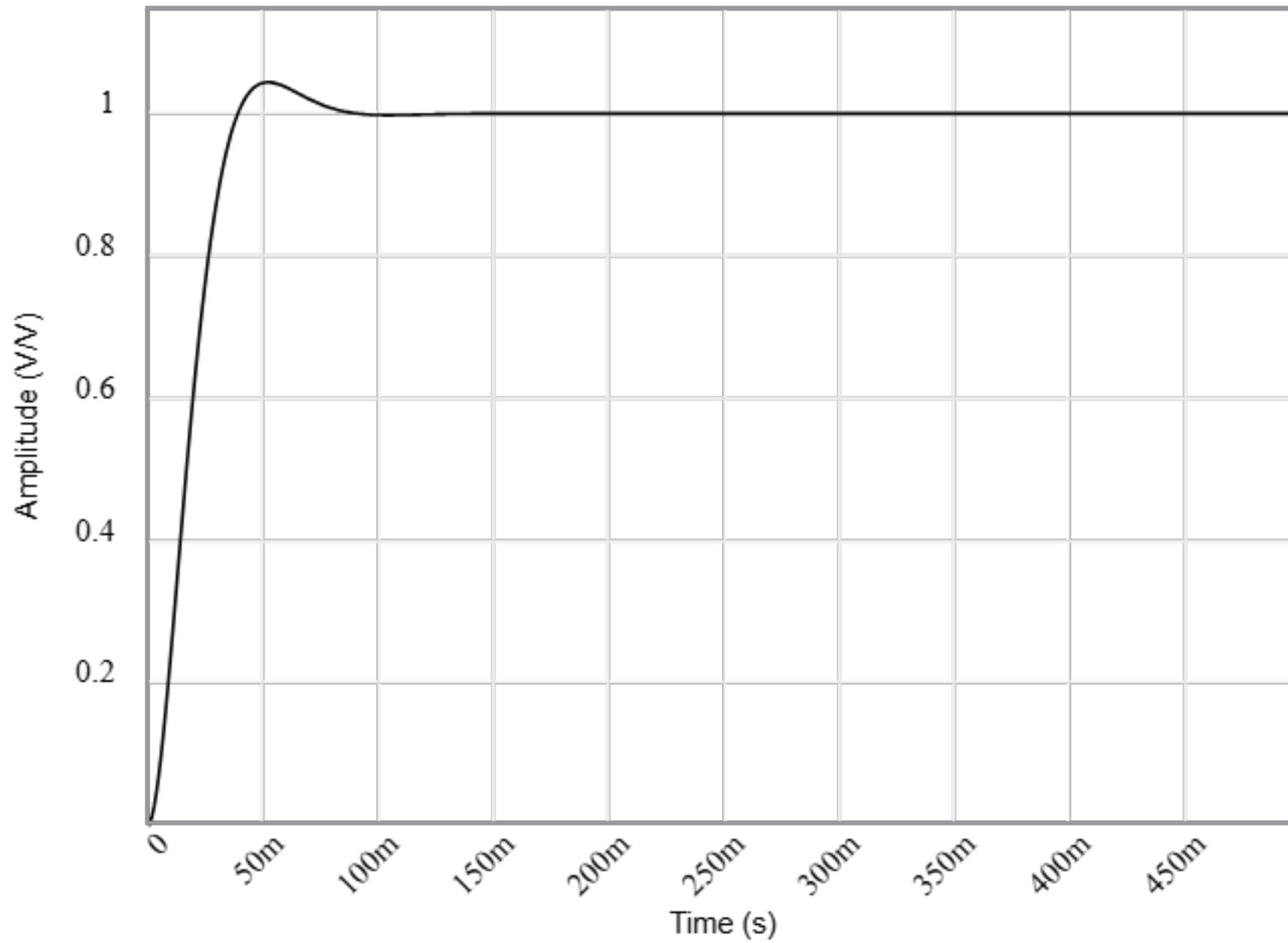
Group Delay



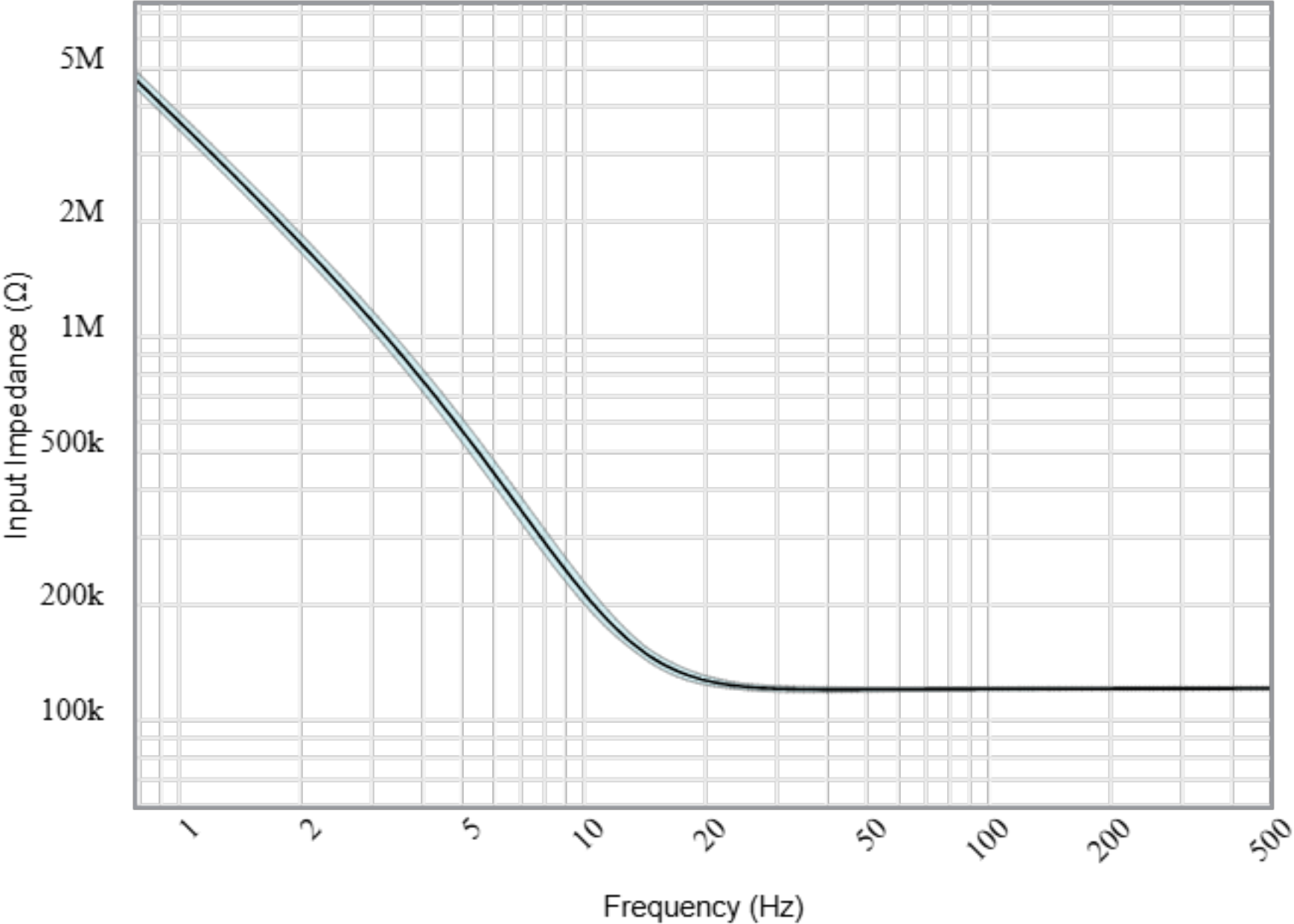
Phase Delay



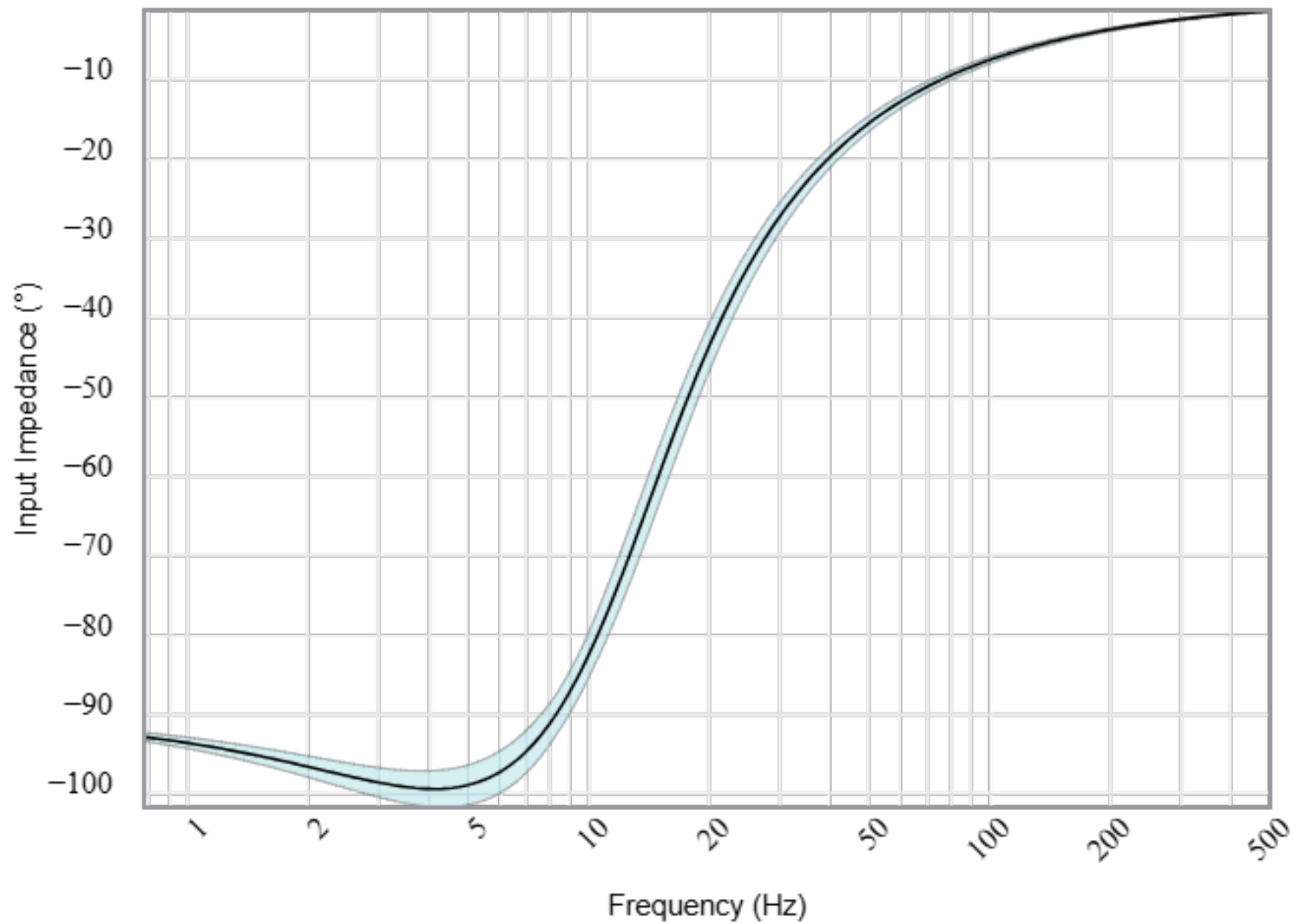
Step Response



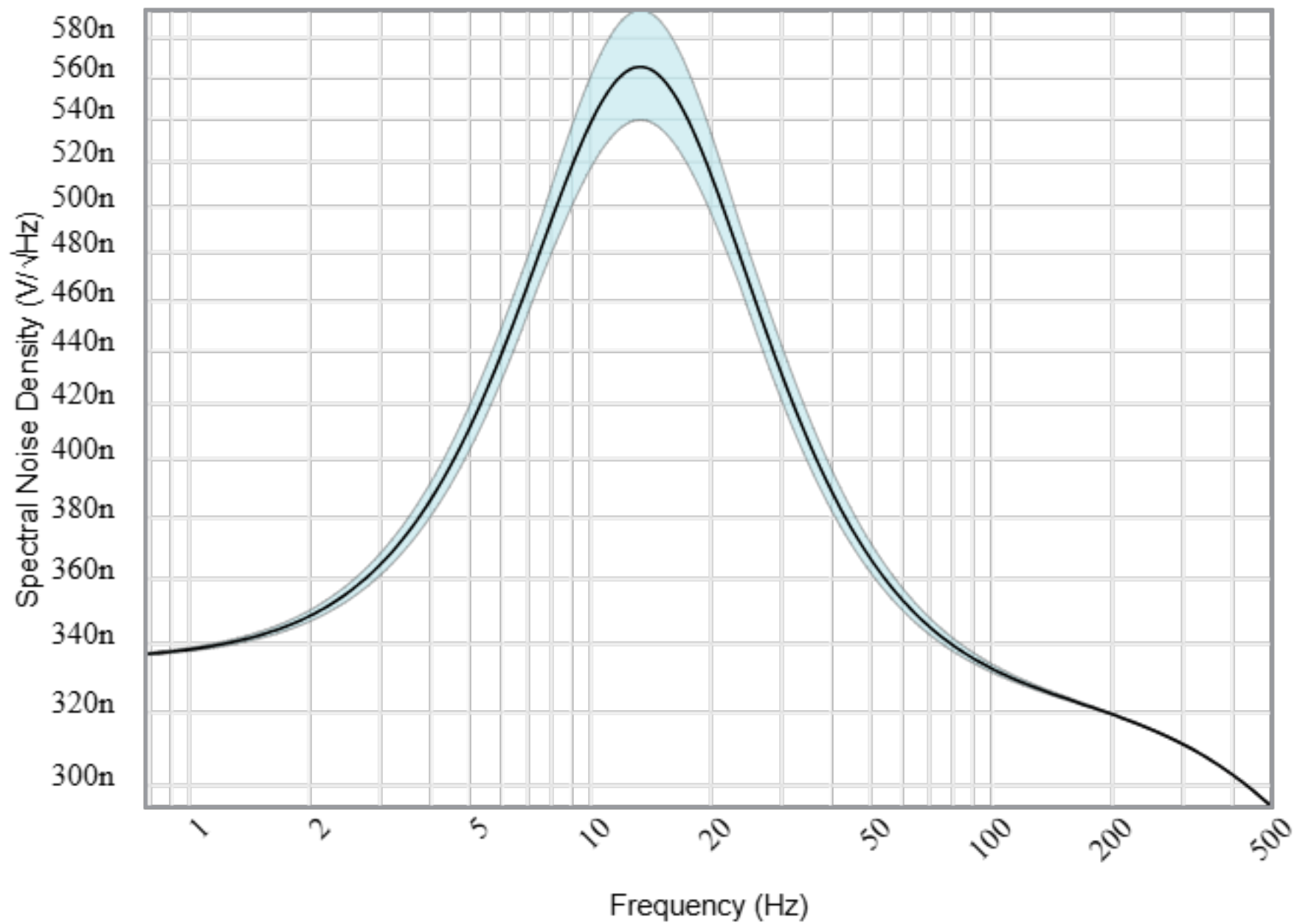
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 1 op amp stage(s) with the following characteristics



2nd order
Low-Pass
Sallen Key

Gain (V/V):

f_p (Hz):

Q:

Target

Simulated

1

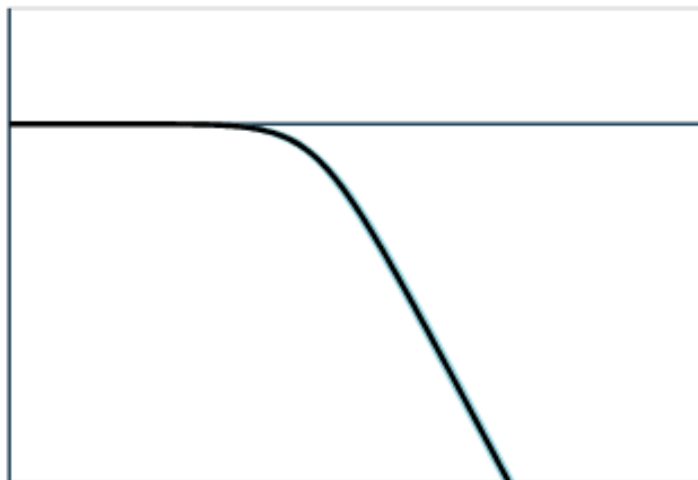
1 to 1

13.5

12.7 to 14.3

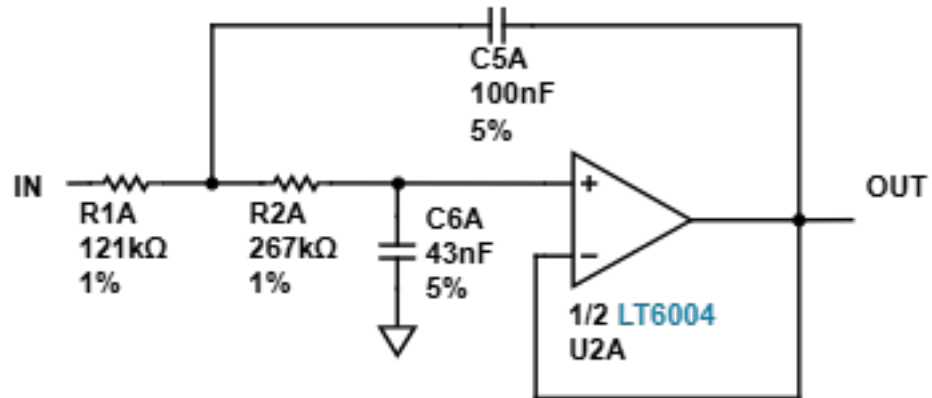
707m

672m to 748m

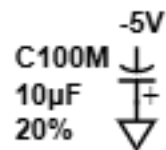
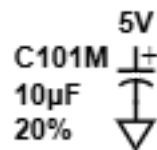
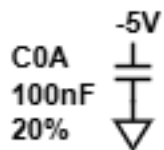
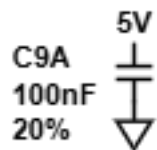


Circuit

Stage A
2nd order
Low-Pass
Sallen Key



BYPASS CAPACITORS



SPARES Why The Spares?

